Toroidal coherence propulsion

What does standard physics predict for the Tier 1 bench of Konstapel's toroidal-coherence paper, and for Hall's reflection reading of lensed galaxies? — register entry H35

kronspice simulation report · 2026-11-07

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Summary

J. Konstapel, "Toroidal Coherence Engineering: From Hall's Photon Theory to Practical Propulsion Systems" (Leiden, 1 Nov 2025, 17 pp.) proposes that heated fiber-optic toroidal coils, phase-locked to the Earth's electromagnetic field, invert a scalar "Sn field" (identified with the scalar part of Maxwell's quaternion equations, after V. Robinson and C. Hall) and reduce the inertial mass of a craft. The paper gives no equation with a value for its coupling σ, so the hypothesis cannot be put into the simulator as a hypothetical model. This example computes the **standard prediction** for each of the paper's three Tier 1 measurements (its §6.1), the ordinary effects a bench must remove, the paper's own figures, and Hall's explanation of lensed galaxy images.

- **The (3n ± 1) harmonic signature is ordinary symmetry.** Three coils at 120° with an ordinary diode nonlinearity, in a kronspice transient, show the multiples of 3 only at the 10^{-10} level; a single coil shows every harmonic.
- **Discrete lines are ordinary loop modes:** a closed 10 m silica loop has longitudinal modes every $c/(n_g \cdot \ell)$ = 20.53 MHz.
- **On the balance, heat makes the coil heavier by E/c² (+0.39 µg),** while air buoyancy (16.7 g) and feed-lead forces (1.3 g per metre) alone reach the claimed 500 mg – 50 g.
- **The paper's figures do not hold:** 1 kN on a 1000 t craft gives 0.001 m/s², not 1 m/s².
- **Hall's reflection at hydrogen clouds turns light by at most 1.3×10^{-3} arcsec** (dense core), against 1–30 arcsec observed in strong lensing, and does so chromatically (2.1 % more for blue than for 1 µm).
- **Status: untested.** Tests proposed: a vacuum balance with a dummy coil at equal power and a written coherence protocol; multiband image positions of a lensed quasar.

The claim and the standard prediction

Tier 1 measurement or claim	Paper (hypothesis)	Standard prediction (this example)
Harmonics of the coil array	a (3n ± 1) "topological signature"	any balanced three-coil array with an ordinary nonlinearity radiates only 3n ± 1; one coil radiates every harmonic
Infrared emission	discrete lines from a coherent coil	longitudinal loop modes every $c/(n_g \cdot \ell)$; thermal light filtered, not phase-locked
Weight on a balance	0.001 %–0.1 % lighter (500 mg – 50 g)	+0.39 µg (E/c²); ordinary forces of mg to g size
Galaxy lensing	frustrated total internal reflection at hydrogen clouds (Hall)	gravitational lensing: achromatic, 1–30 arcsec separations

CLAUDE.md §3c states that the quaternion form of Maxwell's equations is equivalent to the vector form; its scalar part adds no physics. Related entries: H23 (the toroidal electron the paper builds on), H18 (inertial-mass reduction of a craft), H8 and H10 (scalar-field electrodynamics and Bearden's claims, untested).

Part 1 — the (3n ± 1) "topological signature"

Three coils at 120°, each 1 mH + 2 Ω, closed through 20 Ω in parallel with a diode (an ordinary rectifying contact), driven at 10 kHz (inside the paper's 1–100 kHz), solved by kronspice transient. The array field is the space vector $(2/3)(i_a + a \cdot i_b + a^2 \cdot i_c)$, with $a = e^{j2\pi/3}$, the field on the axis of three coils at 120°. Harmonic amplitudes relative to the fundamental:

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n	One coil	Array, forward	Array, backward	Linear control
1	1	1	1.7×10^{-9}	1
2	3.25×10^{-2}	1.5×10^{-10}	3.25×10^{-2}	2.4×10^{-15}
3	1.06×10^{-3}	5.5×10^{-10}	5.3×10^{-10}	1.6×10^{-15}
4	3.12×10^{-3}	3.12×10^{-3}	1.0×10^{-10}	1.0×10^{-15}
5	2.58×10^{-4}	3.0×10^{-10}	2.58×10^{-4}	1.3×10^{-15}
6	8.50×10^{-4}	1.0×10^{-10}	1.2×10^{-10}	1.7×10^{-16}
7	1.01×10^{-4}	1.01×10^{-4}	2.1×10^{-10}	8.0×10^{-16}

- Harmonics 3n + 1 (4, 7) rotate forward, 3n − 1 (2, 5) backward; multiples of 3 cancel by symmetry. The linear control shows no harmonics above round-off.
- Ledger: 11.98 mJ in − 8.65 mJ returned = 10.48 mJ dissipated − 7.16 mJ released from storage (start-up); integration mismatch 1.6 nJ, worst point residual 0.
- The paper's Tier 2 array would show 3n ± 1 with no new physics. For the single Tier 1 coil the only unexpected result would be the *absence* of multiples of 3, which the paper does not state.

Part 2 — discrete infrared lines

A closed 10 m silica loop (group index n_g 1.46, declared) has longitudinal modes every $c/(n_g \cdot \ell)$ = 20.53 MHz; every closed loop shows discrete lines (compare the ring lines of *examples/photon_electron*, AC peaks at $n \cdot v/\ell$). Thermal light at 1000 K peaks at 2.90 μm with a coherence length of order that wavelength, 3×10^{-7} of the loop: the loop filters it into modes but cannot phase-lock it. A test needs a stated line spacing or linewidth that differs from $c/(n_g \cdot \ell)$.

Part 3 — the balance

Rig (declared where the paper is silent): 50 kg on the balance, 100 kW heater (paper), coil at 1000 K (paper), 10 m (paper) × 50 mm bundle, emissivity 0.8, silica $c_p \approx 1000$ J/(kg·K), 250 A DC feed, 50 μT.

Force on the balance	Equivalent mass
Claim: 0.001 %–0.1 % lighter	500 mg – 50 g
Standard prediction: heat adds E/c^2 (heavier)	**+0.39 μg**
Radiation recoil, all emission one way (P/c)	≤ 34 mg
Feed lead in the geomagnetic field	1.3 g per metre
Air buoyancy lost round the hot coil (in air)	16.7 g

- The coil radiates 71 kW at 1000 K, so the heater holds the temperature.
- Ordinary effects reach the claimed band. A bench must run in vacuum, cancel lead forces (twisted, reversed feed), compare against a dummy coil without fiber at the same power, cycle the drive on and off, and fix the "coherence protocol" in writing before the run (the paper attributes null results to poor phase-lock, §5.3, §9.1). A well-shielded balance can then bound the effect below 10^{-6} of the rig mass.
- Existing bound: MICROSCOPE finds no composition-dependent free fall at the 10^{-15} level, η(Ti, Pt) = [−1.5 ± 2.3 (stat) ± 1.5 (syst)]×10^{-15} (Touboul et al., PRL 129, 121102, 2022), which leaves little room for gravity as an electromagnetic coupling.

Part 4 — the paper's figures

- 1 kN on a 1000 t craft gives 0.001 m/s², not 1 m/s² (§4.3); the 5 and 100 m/s² rows inherit the ×1000 error.
- 6000 g on 1000 t with 1 kN needs a residual mass fraction 1.7×10^{-8}, not the ">99.99 %" (10^{-4}) the paper states.
- Coil length: 3–10 km per coil (§4.1), 3,000–5,000 km per coil (§4.3), ~2,000 km in total (§6.3).
- Aluminium cladding (melts at 933 K) is run at 1,500–2,000 K; fused silica softens near 1,940 K (paper: melts at 2,400 K).
- Lock to the 7.8 Hz Schumann mode within 0.1 %: that mode has Q ≈ 5 (Sentman 1995; *examples/earth_machine*), a linewidth of about 20 %.
- $E_{eff} = (1 - f)\cdot\gamma\cdot m_0\cdot c^2$ for v > c (§8.1): γ is not real above c, whatever the mass.

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Part 5 — Hall's "FTIR lensing" of galaxy images

Hall's photon theory explains lensed galaxy images as total (and frustrated total) internal reflection at thin hydrogen clouds with an empty gap between them, not as gravitational lensing. Total reflection from hydrogen to vacuum happens only for rays within the critical grazing angle $\sqrt{(2(n-1))}$ of a sharp boundary, and turns them by at most twice that. With H_2's measured refractivity (Peck & Huang, JOSA 67, 1550, 1977; $n-1 = 1.388\times10^{-4}$ at STP, 589 nm), scaled to cloud density:

H_2 density	$n-1$	Largest deflection
1 cm^{-3} (diffuse)	5.2×10^{-24}	1.3×10^{-6} arcsec
10^4 cm^{-3} (molecular cloud)	5.2×10^{-20}	1.3×10^{-4} arcsec
10^6 cm^{-3} (dense core)	5.2×10^{-18}	1.3×10^{-3} arcsec

- Observed strong-lensing images are separated by about 1–30 arcsec, 10^3–10^7 times more.
- Real cloud edges are graded over parsecs, not the sub-wavelength step that total reflection needs, and frustrated total reflection needs a gap no wider than about a wavelength.
- The reflection is chromatic: blue light is turned 2.1 % more than 1 μm light, while gravitational lensing turns every wavelength equally. **Test:** compare the image positions of a multiply imaged quasar across bands; positions that agree across bands rule out the reflection reading.

Limits

- No hypothetical model is simulated: the paper gives no equation for its coupling σ, so only the standard prediction and the ordinary bench effects are computed.
- The balance rig's mass, bundle size, emissivity, feed current and field are declared where the paper is silent.
- Only Part 1 is a circuit simulation (kronspice transient); Parts 2–5 are closed-form estimates.

Status

Register entry H35 is **untested**: no equation gives the coupling, so there is no hypothetical model to compare. Every claimed Tier 1 signature has an ordinary explanation of at least the claimed size. What would move the entry: a vacuum balance run with a dummy coil at equal power and a coherence protocol fixed in writing beforehand, or lensed-quasar image positions that shift with wavelength.

Reproducing the results

```
go run ./examples/toroidal_coherence
go test ./examples/toroidal_coherence
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Sources: J. Konstapel, "Toroidal Coherence Engineering" (Leiden, 1 Nov 2025); Touboul et al., PRL 129, 121102 (2022); Peck & Huang, JOSA 67, 1550 (1977); Sentman (1995).

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